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1 – OVERVIEW

This section applies to Signa 1.5T systems installed with an SRFD2 Cabinet and explains the procedures for the replacement of the SRFD2 Module.

Parts mentioned are listed in Table 1-1:

TABLE 1-1
FRU(1) PART LIST

Manufacturer's Part Number	GEMS Part Number	Item Description
AN8103	2351573	SRFD2 Module with Case and extra steel rail for Hoist kit (46-317258P1)

WARNING!

PERSONAL INJURY OR PROPERTY DAMAGE. THE SRFD2 MODULE WEIGHS 200 LBS~91 KG. THE UNIVERSAL HOIST KIT IS REQUIRED TO REMOVE/REPLACE THE MODULE FROM THE CABINET.

WARNING!

POSSIBLE PERSONAL INJURY. HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS/33 KG. ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.

Special tools needed to replace the amplifier are listed in Table 1-2:

TABLE 1-2
SPECIAL TOOLS NEEDED

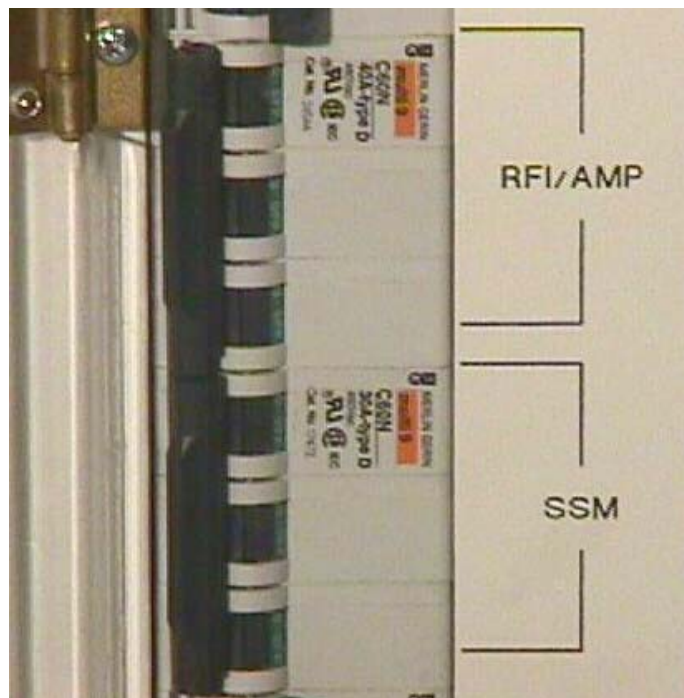
Item	Description	Part Number
1	Universal Hoist Kit – Used to remove amp from cabinet.	46-317266G3, G4, G5 or G6

2 – SRFD2 CABINET POWER LOCKOUT AND TAGOUT



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE SRFD2 CABINET BEFORE ATTEMPTING TO REMOVE THE SRFD2 MODULE. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

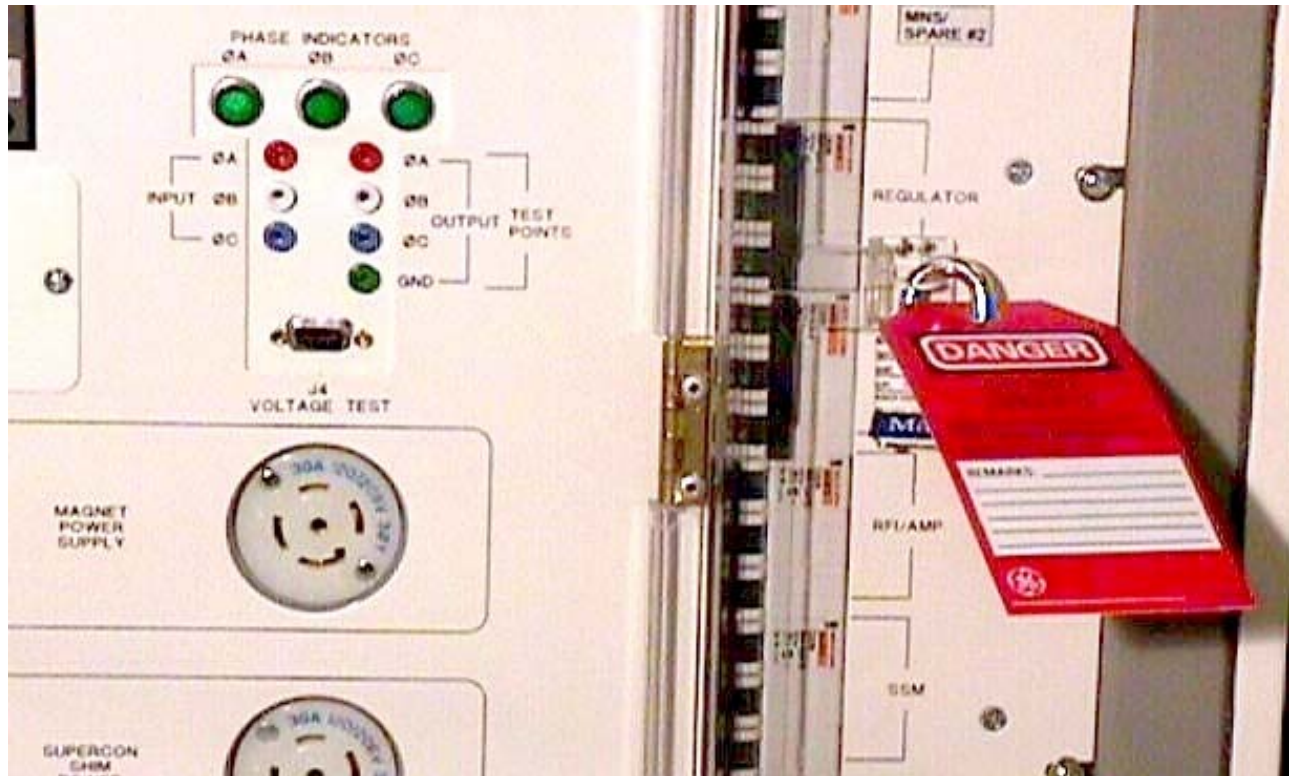
1. Remove the cover from the front of the ACGD cabinet.
2. Open Circuit Breaker Cover on the front of the PDU and switch OFF the **RFI/AMP** and **SSM** breakers. See Illustration 2-1.



BREAKER IDENTIFICATION
ILLUSTRATION 2-1

2 – SRFD2 CABINET POWER LOCKOUT AND TAGOUT (Continued)

3. Close the plexi-glass door and lock and tag it out. See Illustration 2-2. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)



LOCKOUT AND TAGOUT OF BREAKERS
ILLUSTRATION 2-2

4. Remove the front cover from the SRFD2 Cabinet.

2 – SRFD2 CABINET POWER LOCKOUT AND TAGOUT (Continued)

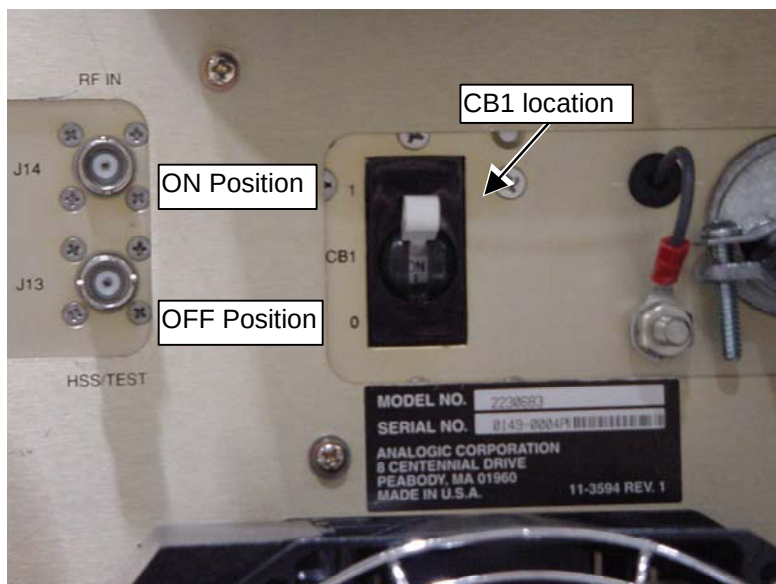
5. Verify all LEDs are OFF (not illuminated) on front of SSM. Verify all LEDs are OFF (not illuminated) on the front of the SRFD2 Module. See Illustration 2-3.



SRFD2 LED LOCATION
ILLUSTRATION 2-3

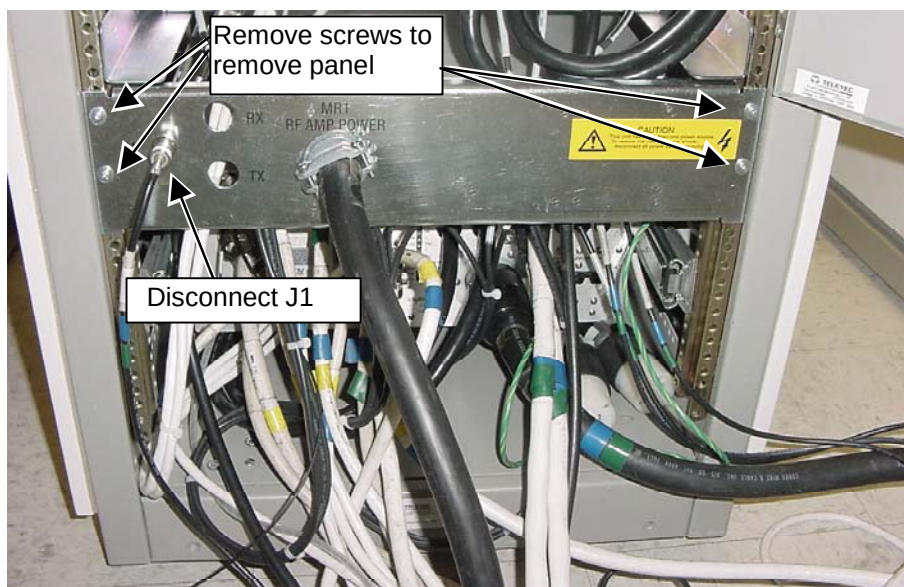
2 – SRFD2 CABINET POWER LOCKOUT AND TAGOUT (Continued)

6. At the rear of the SRFD2 module, place the CB1 breaker in the OFF position (lever down). See Illustration 2-4 for location of breaker.



CB1 LOCATION
ILLUSTRATION 2-4

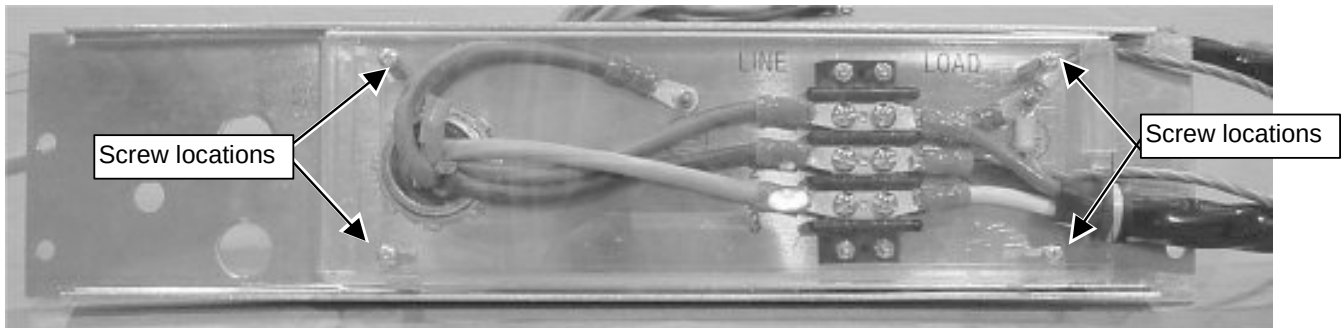
7. Disconnect J1 from the AC Panel from rear of cabinet and remove 4 (four) screws to take panel off cabinet. See Illustration 2-5.



DISCONNECTION LOCATIONS ON AC PANEL
ILLUSTRATION 2-5

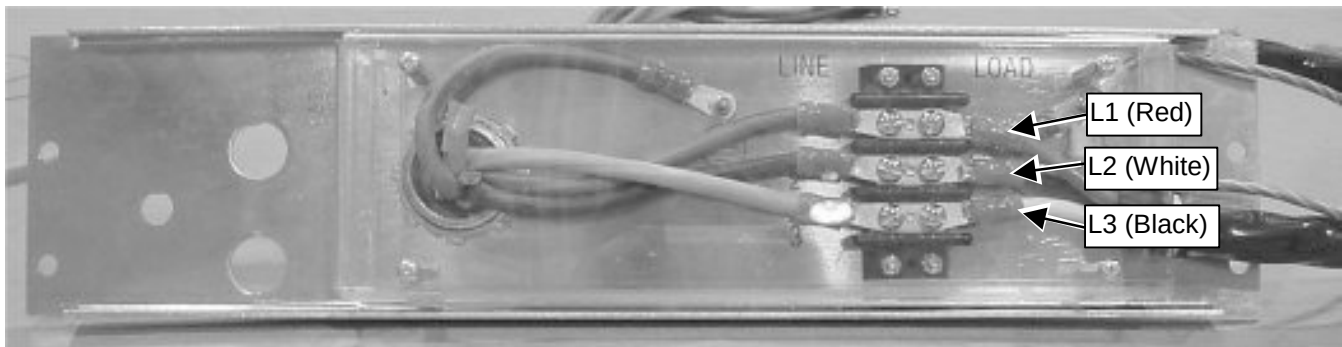
2 – SRFD2 CABINET POWER LOCKOUT AND TAGOUT (Continued)

8. Remove the plastic cover from the AC Panel by loosening the four screws. See Illustration 2-6.



REMOVING PLASTIC COVER ON THE AC PANEL
ILLUSTRATION 2-6

9. Using an AC voltmeter (DVM), measure the input power leads on the rear of the AC Panel TS1 from the rear (right side) of the SRFD2 Cabinet.
 - Measure between L1 and L2; verify voltmeter measures 0 VAC. See Illustration 2-7.



L1 & L2 LOCATIONS
ILLUSTRATION 2-7

3 – SRFD2 AMPLIFIER MODULE

3-1 SRFD2 Amplifier Module Removal

DANGER!!

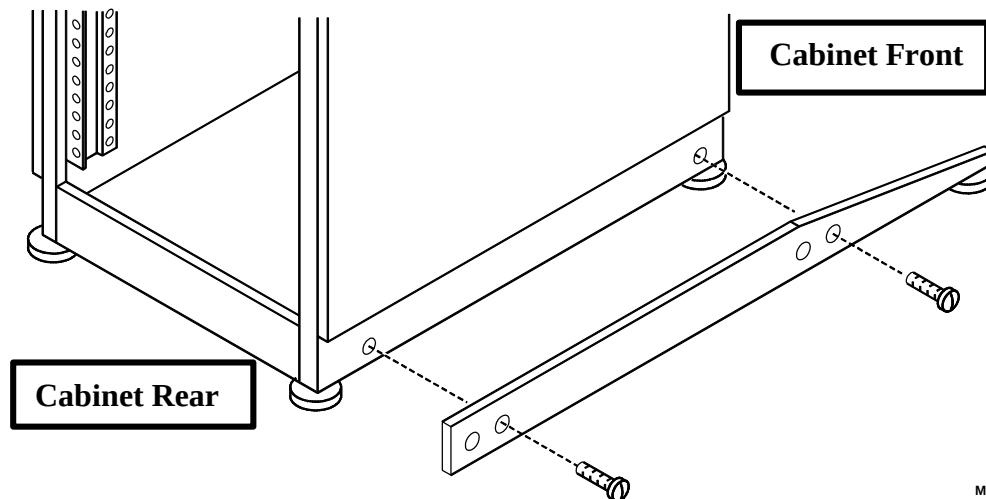
POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE SRFD2 CABINET BEFORE ATTEMPTING TO REMOVE THE RF AMPLIFIER. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

1. Perform Section 2 – **SRFD2 CABINET POWER LOCKOUT AND TAGOUT** (Refer to 8.X Service Methods CD-ROM, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)

DANGER!!

CABINET TIP HAZARD! POSSIBLE PERSONAL INJURY! CONFIRM THAT THE ALUMINUM ANTI-TIP LEGS STORED IN THE REAR OF THE CABINET ARE INSTALLED ON EITHER SIDE OF THE BASE OF THE CABINET BEFORE ATTEMPTING TO REMOVE AN RF AMPLIFIER.

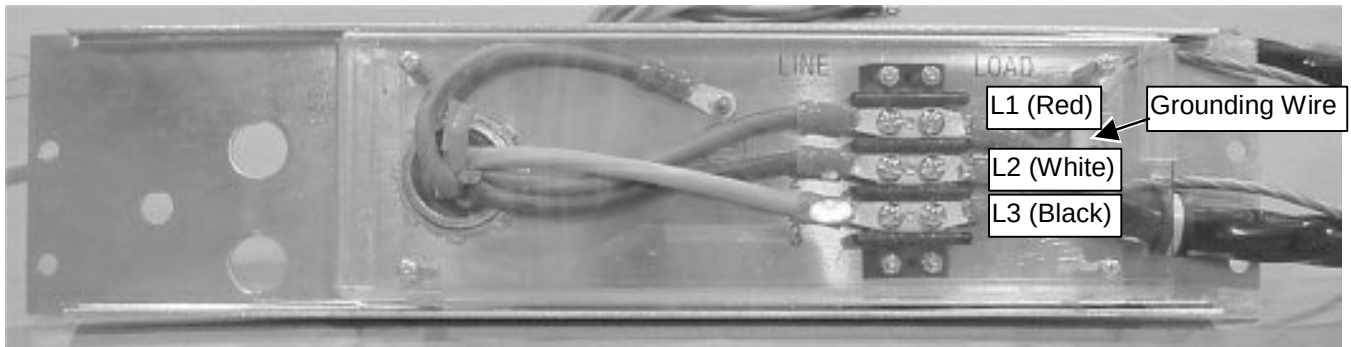
2. Using the screws included with the anti-tip legs, secure the aluminum anti-tip legs to the threaded holes on each side of the base of the SRFD2 Cabinet. See Illustration 3-1.



INSTALLING ANTI-TIP LEGS ON SIDES OF CABINET
ILLUSTRATION 3-1

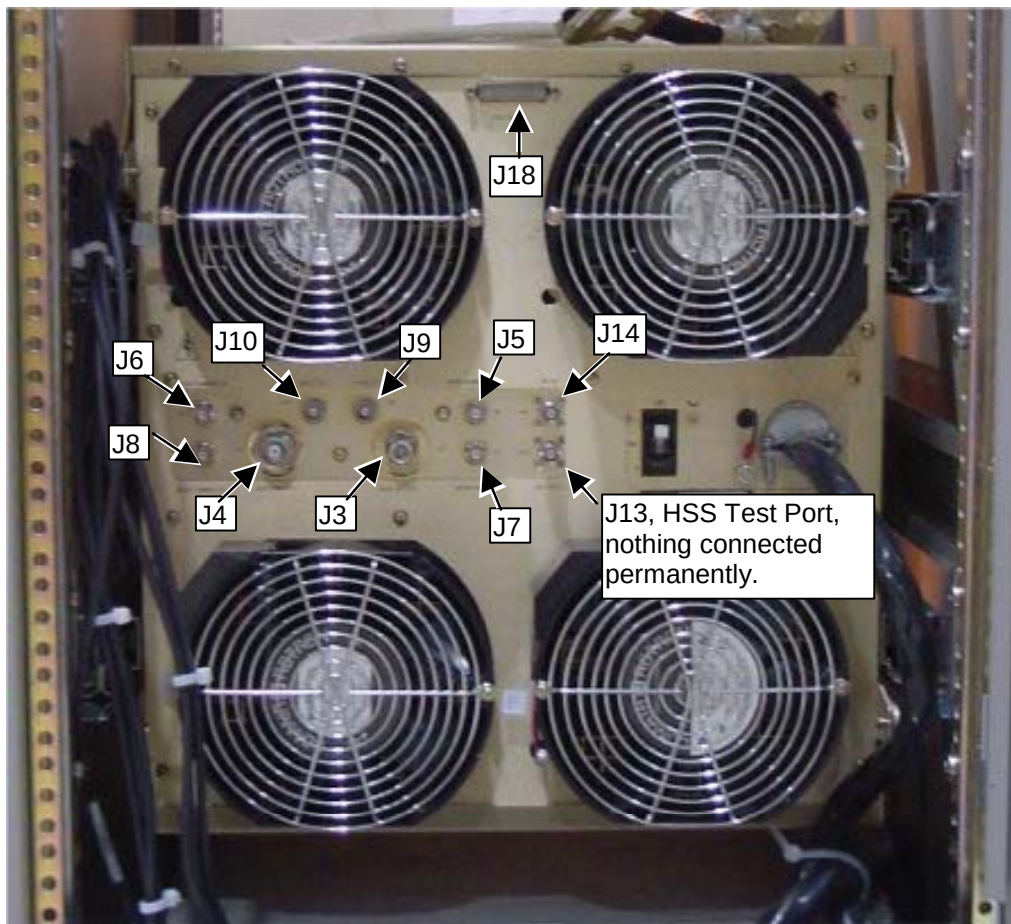
3-1 SRFD2 Amplifier Module Removal (Continued)

3. Disconnect wires from L1, L2, and L3 of LOAD AC connectors and the grounding wire on rear of the AC Panel. See Illustration 3-2.



L WIRE & GROUNDING WIRE LOCATIONS ON AC PANEL
ILLUSTRATION 3-2

4. Remove the J3, J4, J5, J6, J7, J8, J9, J10, J14 and J18 connections on rear of SRFD2 Module. See Illustration 3-3.



CONNECTOR LOCATIONS
ILLUSTRATION 3-3

3-1 SRFD2 Amplifier Module Removal (Continued)

5. Remove the four screws securing the amplifier to the cabinet. See Illustration 3-4.



SCREW LOCATIONS ON CABINET
ILLUSTRATION 3-4

3-1 SRFD2 Amplifier Module Removal (Continued)

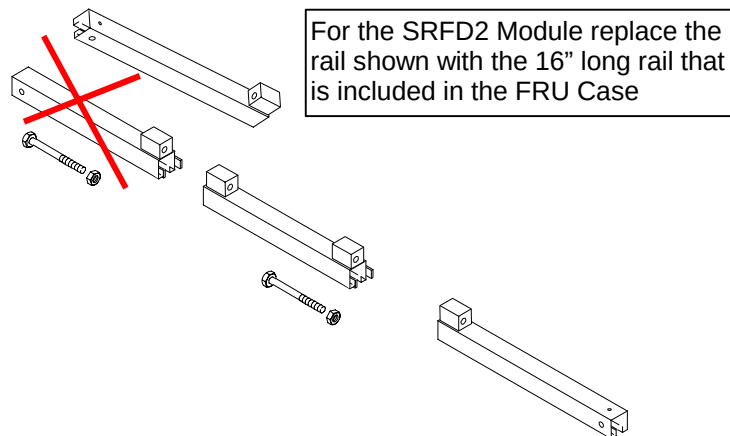
WARNING!

PERSONAL INJURY OR PROPERTY DAMAGE. THE SRFD2 MODULE WEIGHS 200 LBS~91 KG. THE UNIVERSAL HOIST KIT IS REQUIRED TO REMOVE/REPLACE THE MODULE FROM THE CABINET.

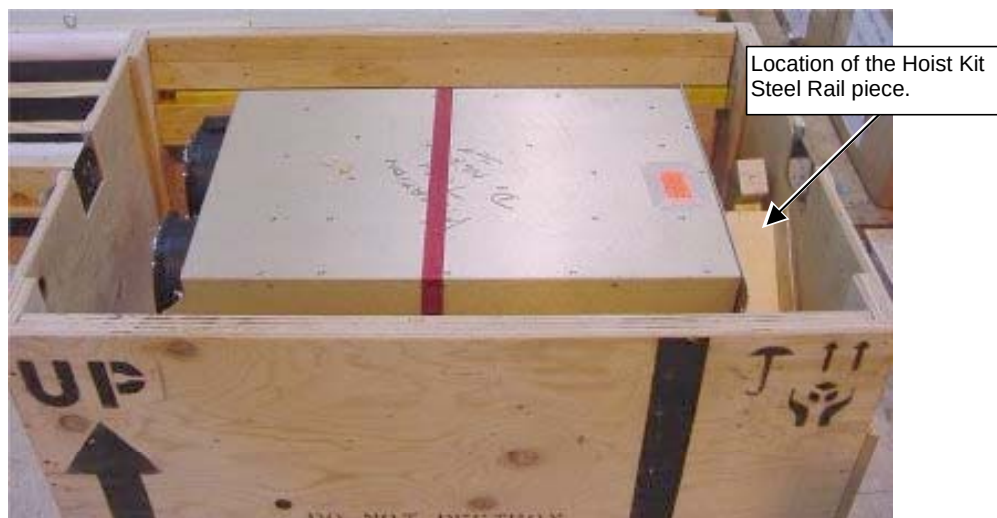
WARNING!

POSSIBLE PERSONAL INJURY. HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS/33 KG. ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.

7. Locate the three (3) large Steel Rail sections and place them on a flat surface for assembly. See Illustration 3-5.



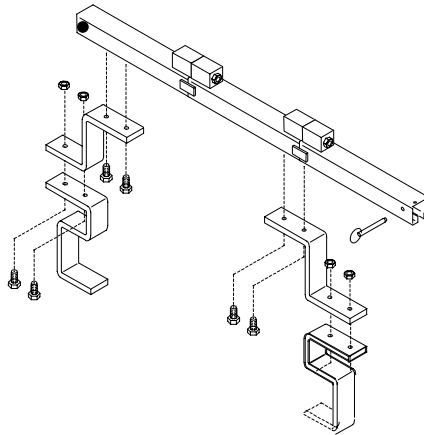
STEEL RAIL SECTIONS, HEX BOLTS, AND HEX NUTS
ILLUSTRATION 3-5



STEEL RAIL LOCATION IN FRU CASE FOR USE WITH HOIST KIT
ILLUSTRATION 3-6

3-1 SRFD2 Amplifier Module Removal (Continued)

8. Arrange the rail sections in the proper order (rear, center, then front) and fasten them together with 4.75" X 1/2" caps crews and hex nuts.
9. Use the crescent wrenches to tighten the cap screws.
10. Assemble and fasten the elevating brackets to the rail assembly. See Illustration 3-7.



HOIST TOP RAIL PARTS WITH ELEVATING BRACKETS
ILLUSTRATION 3-7

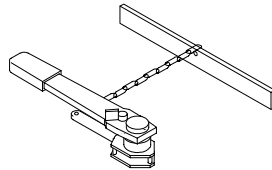
11. Remove the front and rear covers of the SRFD2 Cabinet.
12. Place the Steel Rail and elevating bracket assembly at the top of the SRFD2 Cabinet. Center the Rail Assembly on the top of the RF Amplifier. See Illustration 3-8.



HOISTPARTS REAR VIEW ASSEMBLED
ILLUSTRATION 3-8

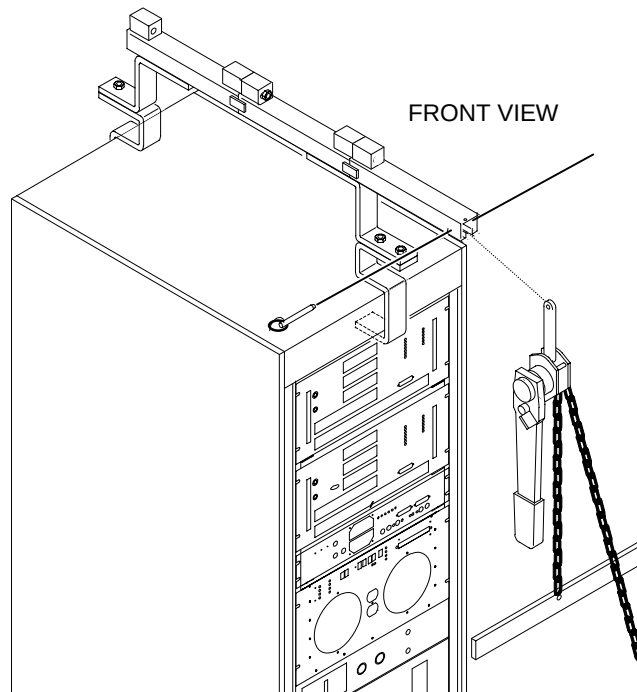
3-1 SRFD2 Amplifier Module Removal (Continued)

13. Locate the Hoist Assembly in the Lift Kit. See Illustration 3-9.



HOIST ASSEMBLY
ILLUSTRATION 3-9

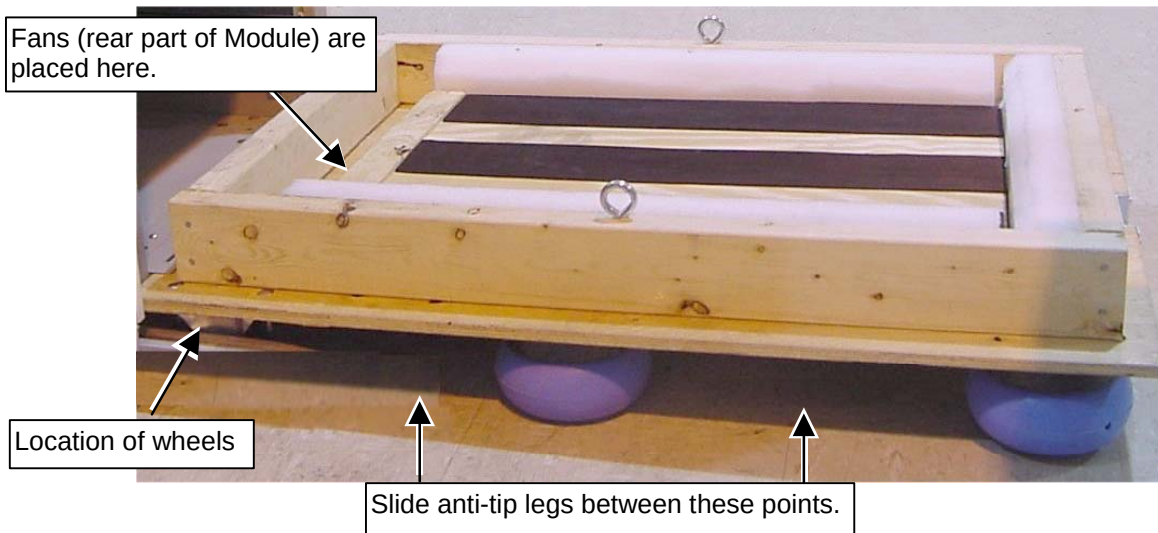
14. Insert the Hoist Assembly into the open end of the front rail. Insert a locking pin into the holes near the front edge of the front rail. This locking pin prevents the Hoist Assembly from sliding out of the rails. See Illustration 3-10.



PLACEMENT OF HOIST ASSEMBLY INTO FRONT END OF STEEL RAIL ASSEMBLY
ILLUSTRATION 3-10

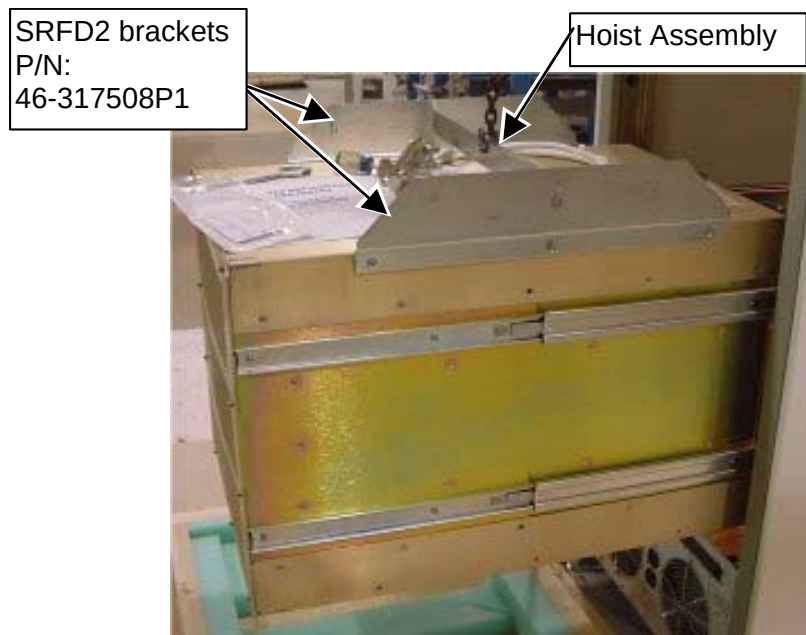
3-1 SRFD2 Amplifier Module Removal (Continued)

15. Remove the top from the FRU case by releasing the butterfly clamp on each side. The top is placed in front of the SRFD2 cabinet for the failed module to be placed on, becoming the bottom of the case for returning the failed module. See Illustration 3-11.



POSITION OF SHIPPING CASE BASE FOR REMOVAL/REPLACEMENT OF MODULE
ILLUSTRATION 3-11

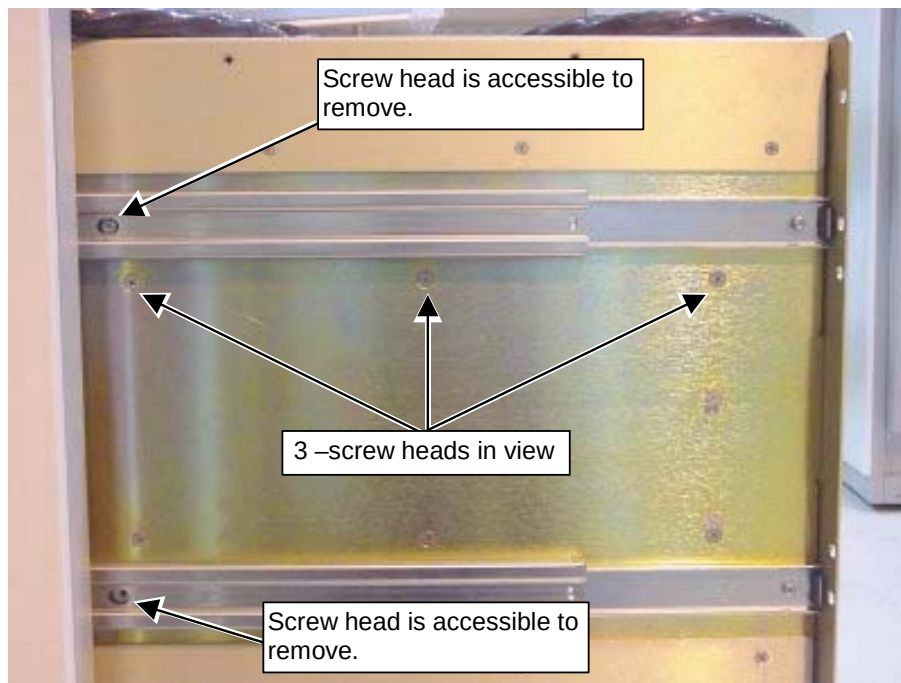
16. Select the proper lift brackets (P/N 46-317508P1) and attach to side of module, then attach the Hoist Assembly to the lift bracket. See Illustration 3-12. Reuse the screws from the front of the cabinet to secure the panels.



ATTACHING LIFT BRACKET TO HOIST ASSEMBLY
ILLUSTRATION 3-12

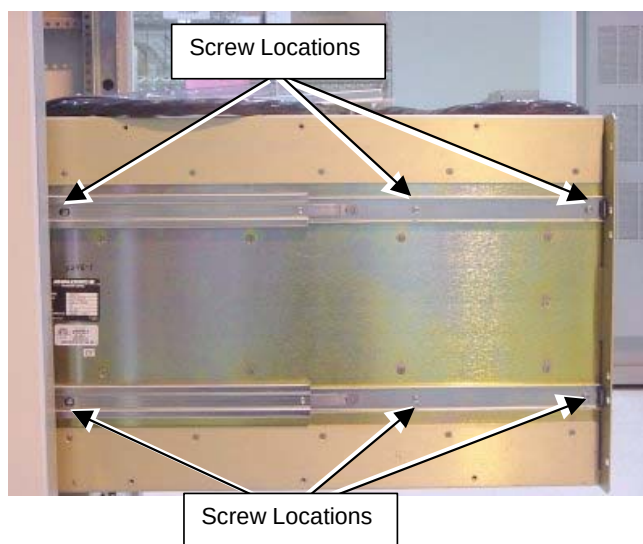
3-1 SRFD2 Amplifier Module Removal (Continued)

17. Remove 4 (four) screws from each sliding rail; however, one screw is not accessible when the rails are fully extended. See Illustration 3-13 and Illustration 3-14.
 - a. Pull the module out of the cabinet as shown in Illustration 3-13, to the point that only three screw heads on the side of the module are in view and to the point where the hole in the rail provides access to the screw. Remove screws from all rails.



ACCESSING RAIL SCREW
ILLUSTRATION 3-13

- b. Pull module out to point where rails are fully extended and remove the 3 remaining screws from each rail. See Illustration 3-14.



LOCATION OF RAIL SCREWS
ILLUSTRATION 3-14

3-1 SRFD2 Amplifier Module Removal (Continued)

18. Push the slides all the way into the cabinet, turn the module so the fans are placed properly on the FRU case and lower the module.
19. Remove the Hoist brackets from the module.

3-2 SRFD2 Amplifier Installation

1. Place the new module that is on the FRU case base in front of the cabinet by sliding base between the anti-tip legs and remove sidewalls by releasing the butterfly clamp. See Illustration 3-15.



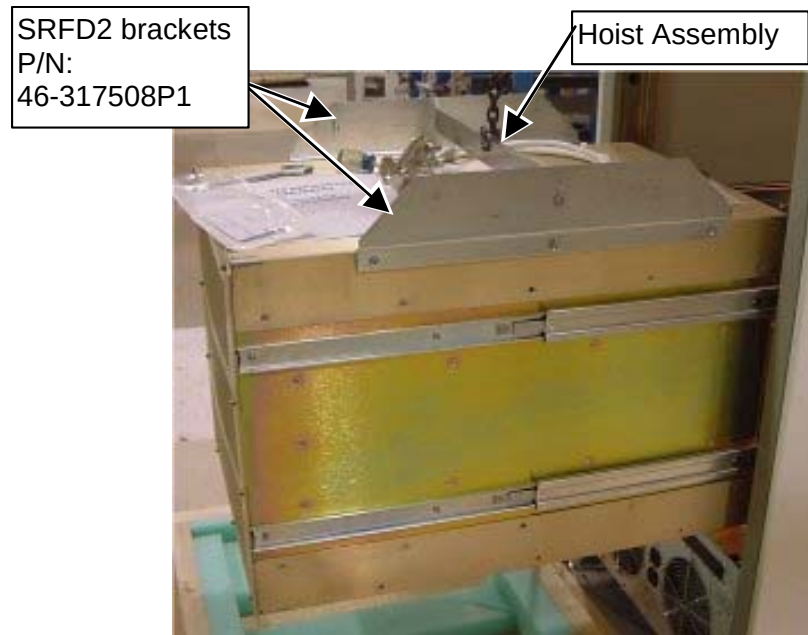
Slide the anti-tip legs between these points.

SIDEWALLS REMOVED FROM FRU CASE
ILLUSTRATION 3-15

2. Remove the shipping strap.
3. Attach the hoist assembly to the hoist brackets.

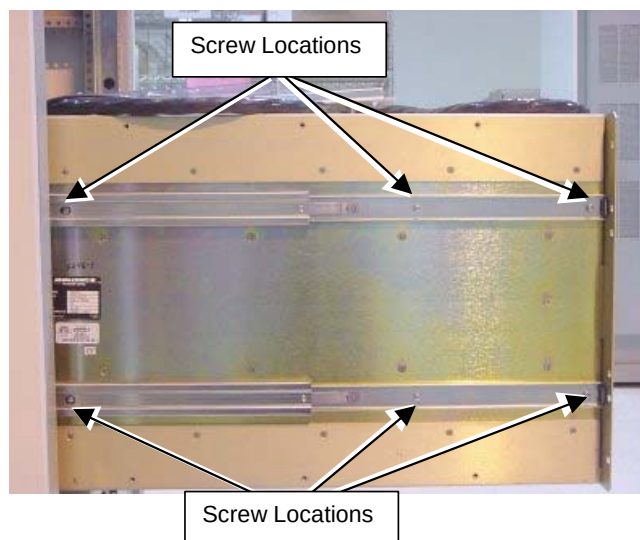
3-2 SRFD2 Amplifier Installation (Continued)

4. Attach the brackets to the SRFD2 module. See Illustration 3-16.



ATTACHING LIFT BRACKET TO HOIST ASSEMBLY
ILLUSTRATION 3-16

5. Using the hoist, raise the module to align with attaching the rails.
6. Once the module is at the correct height to attach the rails, slide the rails out completely and secure the 3 screws that are accessible. See Illustration 3-17.

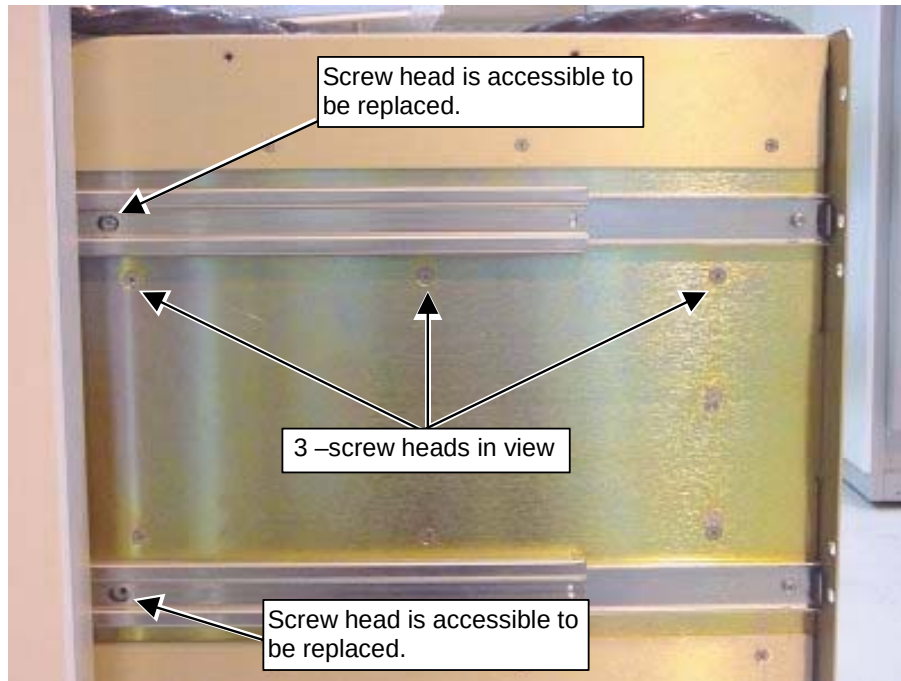


LOCATION OF RAIL SCREWS
ILLUSTRATION 3-17

7. Remove the hoist brackets from the module and return to the Hoist Kit.

3-2 SRFD2 Amplifier Installation (Continued)

8. Push the module in to gain access to the fourth screw on each rail and secure rail by replacing the screw. See Illustration 3-18.



ACCESSING RAIL SCREW
ILLUSTRATION 3-18

3-2 SRFD2 Amplifier Installation (Continued)

9. Slide module completely in cabinet and insert four screws into front of amplifier to secure to cabinet. See Illustration 3-19.



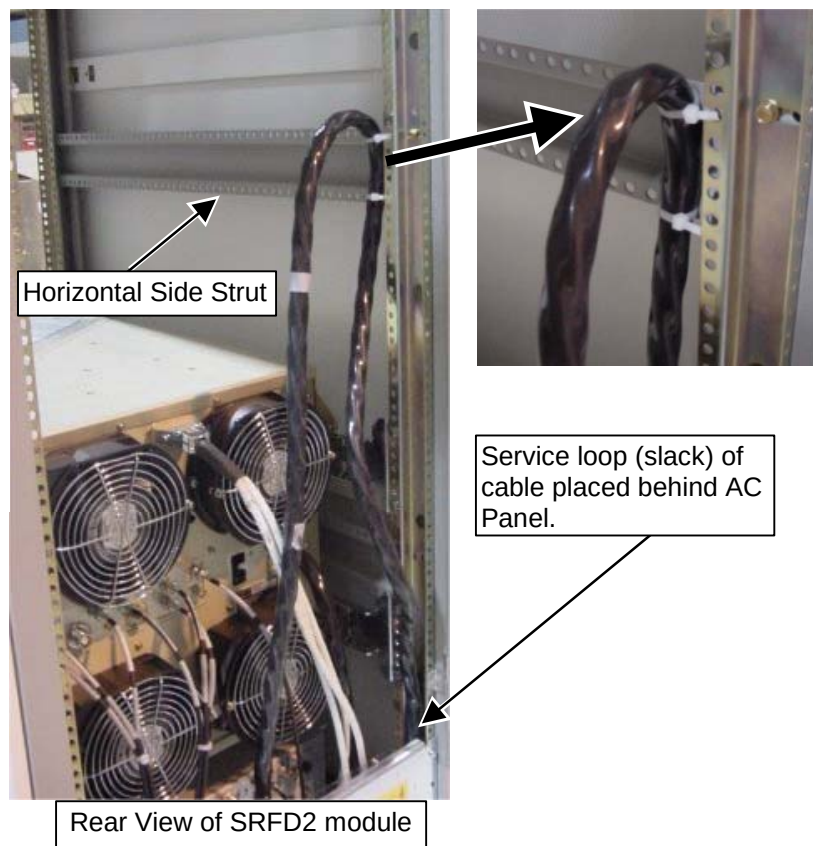
SCREW LOCATIONS ON CABINET
ILLUSTRATION 3-19

3-2 SRFD2 Amplifier Installation (Continued)

WARNING!

SHOCK HAZARD! TO PREVENT THE POSSIBLE ELECTRIC SHOCK FROM AN EXPOSED POWER CABLE, DO NOT FEED THE SRFD2 PIGTAIL/POWER CABLE BEHIND THE SLIDING RAILS. THIS MAY CAUSE DAMAGE TO THE CABLE POTENTIALLY EXPOSING CABLES. SEE STEP 9 AND ILLUSTRATION 3-20 FOR THE CORRECT CABLE DRESSING.

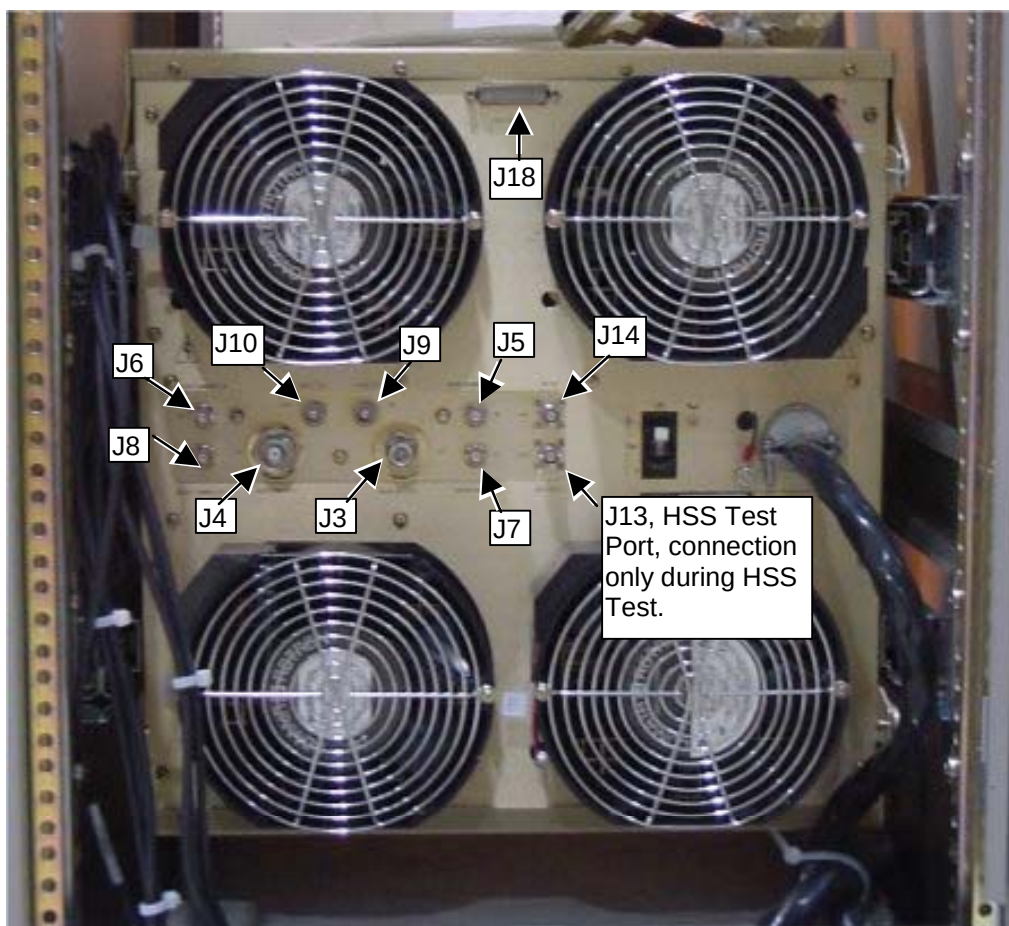
9. Dress the pigtail cable to provide a service loop (1020 mm/40.15 inches) for the SRFD2 module. Attach the cable to the upper horizontal side strut in cabinet using ty-wraps; see Illustration 3-20. It is important that the cable service loop is placed behind the AC Panel and behind the module. **DO NOT** place the cable on the side of the cabinet, behind the sliding rails. This will cause the cable to become damaged and create a hazard.



DRESSING THE PIGTAIL CABLE IN THE SRFD2 CABINET
ILLUSTRATION 3-20

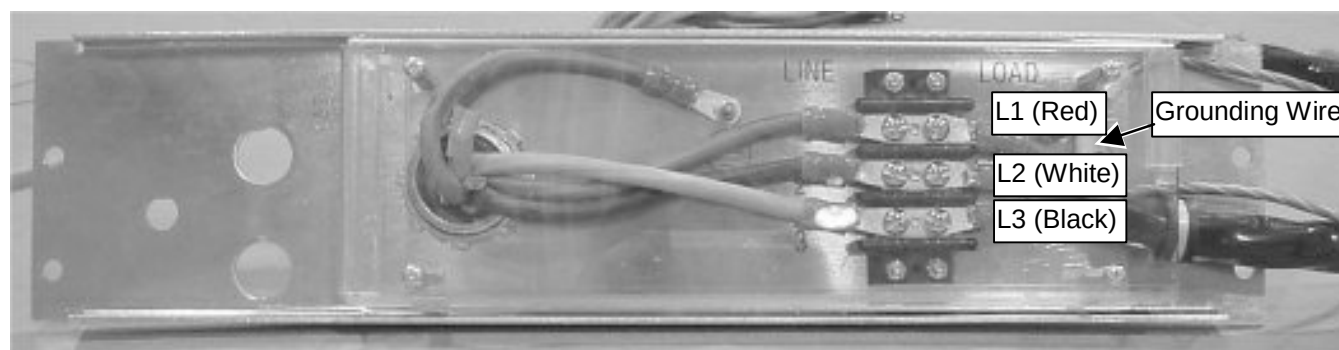
3-2 SRFD2 Amplifier Installation (Continued)

10. Reconnect the J3, J4, J5, J6, J7, J8, J9, J10, J14 and J18 connections on rear of SRFD2 Module. See Illustration 3-21.



CONNECTOR LOCATIONS
ILLUSTRATION 3-21

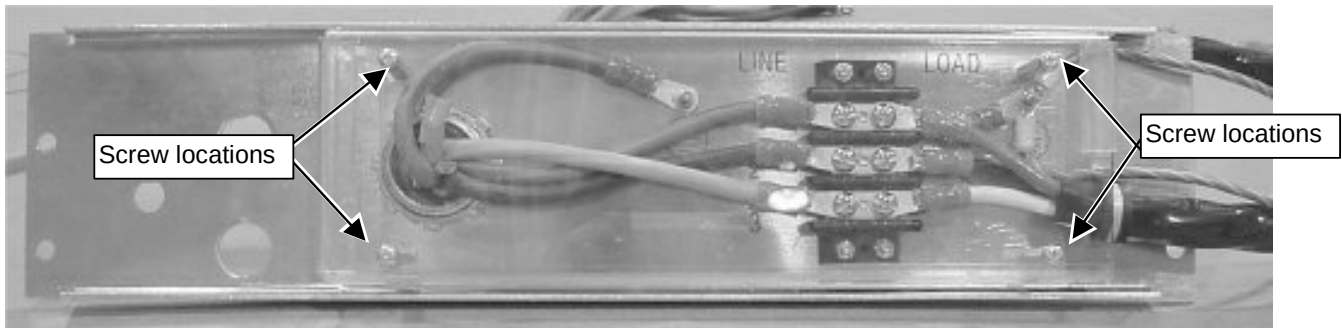
11. Reconnect the L1, L2, and L3 of LOAD AC connectors and the grounding wire on rear of the AC Panel. See Illustration 3-22.



L WIRE & GROUNDING WIRE LOCATIONS ON AC PANEL
ILLUSTRATION 3-22

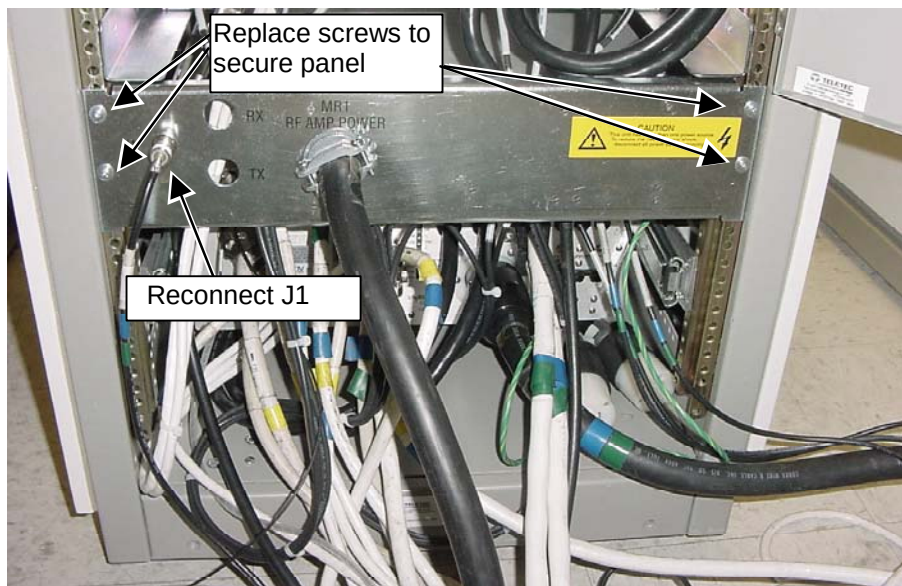
3-2 SRFD2 Amplifier Installation (Continued)

12. Replace the plastic cover on the AC Panel by securing the four screws. See Illustration 3-23.



REPLACING PLASTIC COVER ON THE AC PANEL
ILLUSTRATION 3-23

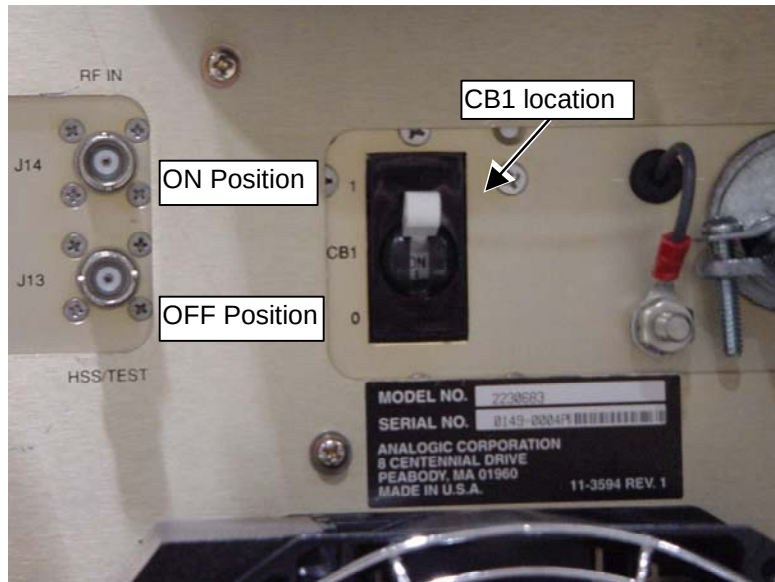
13. Replace the 4 (four) screws to secure the AC Panel to the rear of the cabinet and reconnect J1. See Illustration 3-24.



REAR OF CABINET—AC PANEL
ILLUSTRATION 3-24

3-2 SRFD2 Amplifier Installation (Continued)

14. At the rear of the SRFD2 module, place the CB1 breaker in the ON position (lever up). See Illustration 3-25 for location of breaker.



CB1 LOCATION
ILLUSTRATION 3-25

15. Remove the Universal Hoist from the cabinet.
16. Close the SRFD2 rear cabinet door.
17. Remove lock and tag from the Circuit Breaker cover on the front of the PDU cabinet and put the RFI/AMP and the SSM breakers in the ON position.
18. Confirm that LEDs are illuminated on the front of the SRFD2 Module and SSM and confirm that the fans at the rear of the SRFD2 Module are rotating.
19. If the SRFD2 Module was replaced then perform the calibration procedure described in **1.5T SRFD2 Max Power RF Out Setup and Calibration** (RC7SCA1.DOC) on the 8.X Service Methods CDROM.

3-3 Returning Failed Module

Prepare the failed module by performing the following steps.

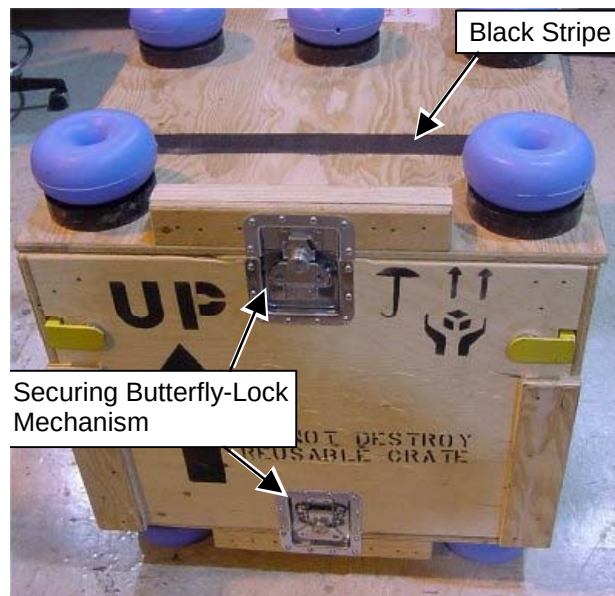
1. Secure the module with the shipping strap and wrap the pigtail power cord tucked on side of the module inside the shipping strap. See Illustration 3-26.



MODULE PREPARED FOR RETURN SHIPMENT
ILLUSTRATION 3-26

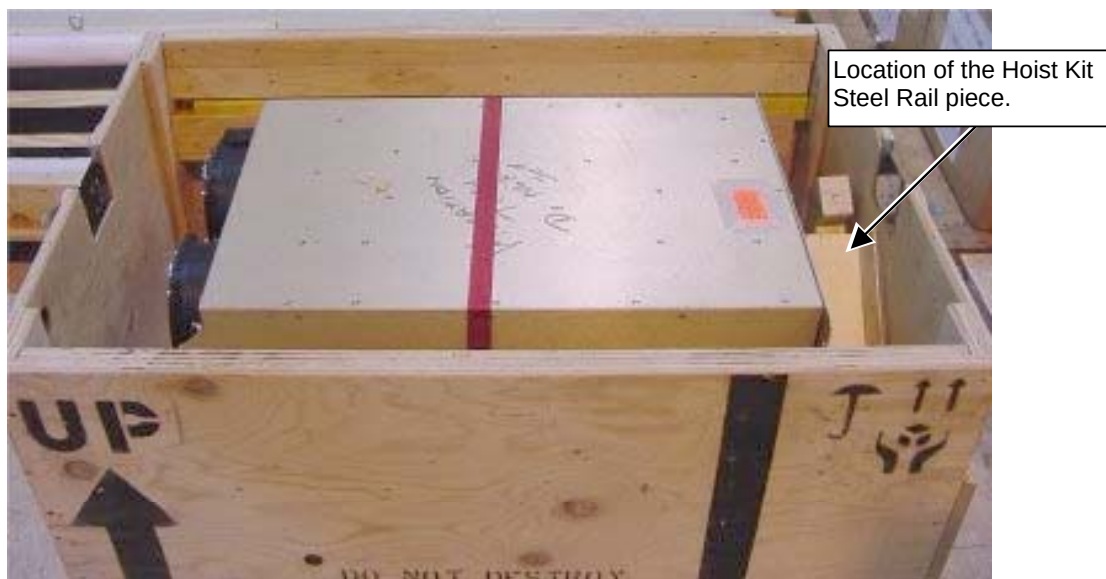
3-3 Returning Failed Module (Continued)

2. Place sidewalls on base and secure. The black stripe on the outside of the box should align with the black stripe on the base that the module resides. See Illustration 3-27.



SECURING HARDWARE LOCATION ON FRU CASE
ILLUSTRATION 3-27

3. Return the 16-inch Hoist rail to the FRU case. See Illustration 3-28.



STEEL RAIL LOCATION IN FRU CASE FOR USE WITH HOIST KIT
ILLUSTRATION 3-28

4. Place top on box, remember to keep the black stripe aligned and secure. See Illustration 3-27.
5. Return module for root cause analysis by vendor.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	April 2, 2002	K. Keshena	Preliminary Release
0	July 30, 2002	K. Keshena	Review for Initial Release
1	Oct.31, 2002	K. Keshena	Corrected the FRU part number and corrected the wiring labels for the AC Panel.
2	10/17/2007	K.Keshena	PSR 13105722, requires the warning statement for the Hoist Kit.